

VIRGEN, Austria

WMO: 112520

Lat: 47.003N

Lon: 12.456E

Elev: 3914

StdP: 12.73

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	13.6	-0.2	6.3	13.2	3.3	7.5	17.4	9.9	29.2	8.5	29.3	2.4	270	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	79.5	62.1	76.4	60.9	73.1	59.6	64.0	75.1	62.5	72.8	61.0	70.3	5.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	91.4	66.8	59.0	86.4	65.1	57.5	81.6	64.2	31.4	75.6	30.2	73.0	29.1	70.3	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.2	7.5	DB	7.8	84.4	4.9	3.2	4.3	86.7	1.4	88.6	-1.4	90.4	-5.0	92.7	(4)
(5)			WB	6.2	66.5	4.7	1.9	2.7	67.9	0.0	69.0	-2.7	70.0	-6.2	71.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	27.7	30.6	37.3	44.1	51.7	58.1	61.3	60.3	53.0	46.1	36.6	30.2		
	DBStd	13.46	7.01	7.71	7.63	6.88	6.12	6.63	5.59	5.92	5.74	6.35	6.94	7.37		
	HDD50	3097	691	542	396	198	54	11	3	34	151	403	613			
	HDD65	7423	1156	962	859	628	415	221	140	168	359	586	851	1078		
	CDD50	1210	0	0	2	20	105	253	351	323	125	29	2	0		
	CDD65	63	0	0	0	0	1	14	25	23	0	0	0	0		
	CDH74	540	0	0	0	0	16	143	218	159	4	0	0	0		
	CDH80	69	0	0	0	0	0	16	29	24	0	0	0	0		
(14) Wind	WSAvg	2.9	2.4	2.6	3.2	3.6	3.6	3.5	3.1	2.9	2.5	2.4	2.3	2.3		
(15) Precipitation	PrecAvg	40.9	1.7	1.4	2.1	2.5	4.0	5.4	6.1	5.9	3.9	3.6	2.8	1.7		
	PrecMax	50.9	3.4	3.7	6.5	5.0	5.7	7.5	8.7	8.8	6.0	7.1	6.5	4.2		
	PrecMin	34.7	0.4	0.1	0.5	0.1	1.6	3.1	2.2	3.7	1.9	0.3	0.0	0.0		
	PrecStd	3.9	0.9	0.9	1.3	1.3	1.3	1.3	1.5	1.3	1.1	1.7	1.5	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	54.1	61.1	68.1	76.6	82.1	83.8	83.2	73.7	65.1	56.7	51.1		
		MCWB	39.4	42.9	46.9	53.1	58.7	61.9	64.0	64.2	60.1	54.6	46.7	40.8		
	2%	DB	43.2	48.5	56.8	63.1	70.9	78.2	79.2	78.6	69.2	61.3	51.2	44.8		
		MCWB	36.4	39.4	45.0	49.8	56.1	61.0	62.1	62.6	57.7	52.3	44.2	37.0		
	5%	DB	39.7	44.1	52.7	59.5	67.1	74.6	76.1	74.6	65.9	58.2	47.8	41.1		
		MCWB	34.3	36.5	42.6	47.8	54.0	59.3	60.9	61.1	56.3	50.8	42.9	35.9		
	10%	DB	36.6	40.5	48.5	55.8	63.2	70.7	72.5	71.1	62.6	54.9	45.2	38.6		
		MCWB	32.2	34.5	40.2	45.8	52.0	57.6	59.4	59.7	54.8	49.4	41.6	34.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	43.5	48.4	53.9	60.0	64.7	65.8	66.4	61.6	56.4	48.6	41.8		
		MCDB	45.7	52.5	59.4	66.9	72.9	75.2	78.9	77.1	69.8	62.1	53.3	47.9		
	2%	WB	37.3	39.9	45.6	50.8	57.4	62.3	63.9	64.1	59.5	53.8	45.8	38.8		
		MCDB	41.4	47.5	55.2	61.4	68.5	74.8	75.1	75.4	66.2	59.1	49.0	42.8		
	5%	WB	35.0	37.3	43.3	48.5	55.3	60.6	62.5	62.5	57.6	51.9	44.1	36.8		
		MCDB	38.6	42.7	51.6	57.9	64.9	71.5	72.5	71.9	63.4	56.4	46.7	40.1		
	10%	WB	33.0	35.1	41.0	46.5	53.1	58.8	61.0	60.8	55.9	50.2	42.1	34.8		
		MCDB	35.9	39.6	47.4	54.3	61.0	68.6	69.8	69.1	61.1	54.3	44.8	37.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.9	15.6	16.1	16.6	17.3	17.7	16.5	15.3	13.3	10.2	9.6		
		MCDBR	13.5	17.9	21.4	22.2	23.0	23.9	23.1	22.1	19.8	17.1	13.4	12.3		
		MCWBR	9.8	12.0	12.8	11.8	10.9	10.0	9.4	9.3	10.7	10.4	9.5	9.0		
		MCDBR	11.8	16.2	20.2	20.8	21.2	21.7	20.2	20.0	16.9	15.3	11.7	11.1		
	5% WB	MCWBR	9.1	11.3	12.4	11.4	10.5	9.6	9.3	9.3	9.7	9.9	8.7	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.280	0.326	0.371	0.386	0.395	0.391	0.385	0.350	0.324	0.290	0.256		
	taud		2.353	2.271	2.237	2.237	2.271	2.290	2.303	2.339	2.413	2.457	2.479	2.425		
	Ebn at Noon		270	282	283	279	279	276	275	272	273	264	254	257		
	Edh at Noon		22	30	37	41	41	41	40	36	31	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		442	725	1059	1325	1528	1705	1737	1532	1136	774	445	367		
	RadStd		51	94	128	173	185	157	148	166	141	88	78	53		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (87 neighbors)		+0.77	N/A	N/A	N/A	+0.74	+0.68	N/A	-214	+131	+54

Nomenclature: See separate page